

End-to-End Solution for 3.5D Style Designs with Stacked Die on Advanced Packages

CT Kao, Product Engineering Group Director, Digital Signoff Group

April 16, 2026

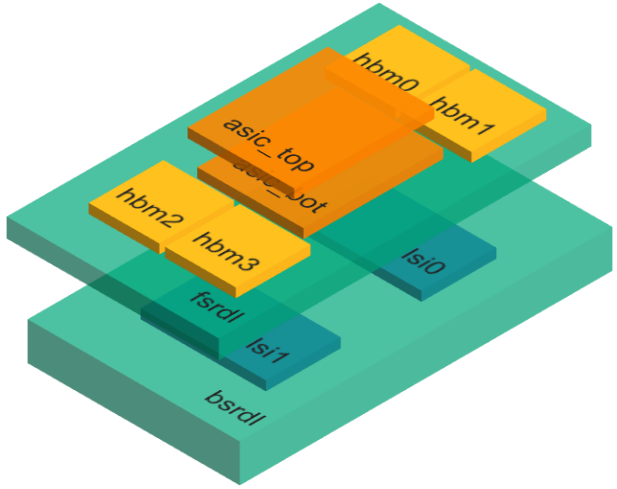
cadence®

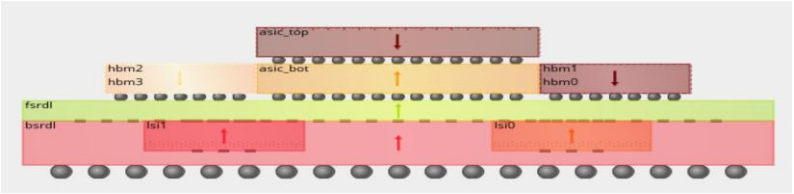
Statements in this presentation (or statements made by Cadence elsewhere that are referenced in this presentation) that refer to plans or expectations are forward-looking statements. These statements are based on management's expectations as of the date hereof and involve many risks, uncertainties and assumptions that could cause actual results to differ materially from those expressed or implied in such statements. For more information on the factors that could cause actual results to differ materially, please refer to the filings we make with the U.S. Securities and Exchange Commission, including our most recent report on Form 10-K, subsequent reports on Form 10-Q and other future filings. Except as required by law, we disclaim any obligation to update these forward-looking statements to reflect future events or circumstances.

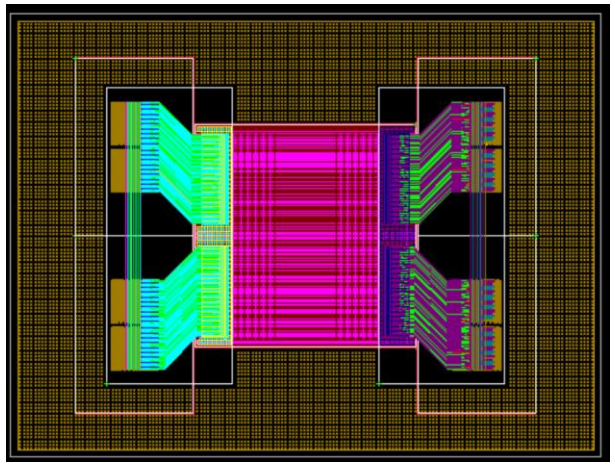
© 2026 Cadence Design Systems, Inc. All rights reserved.

3.5D Design and Analysis RAK

Stacking dies on RDL interposer with silicon bridges



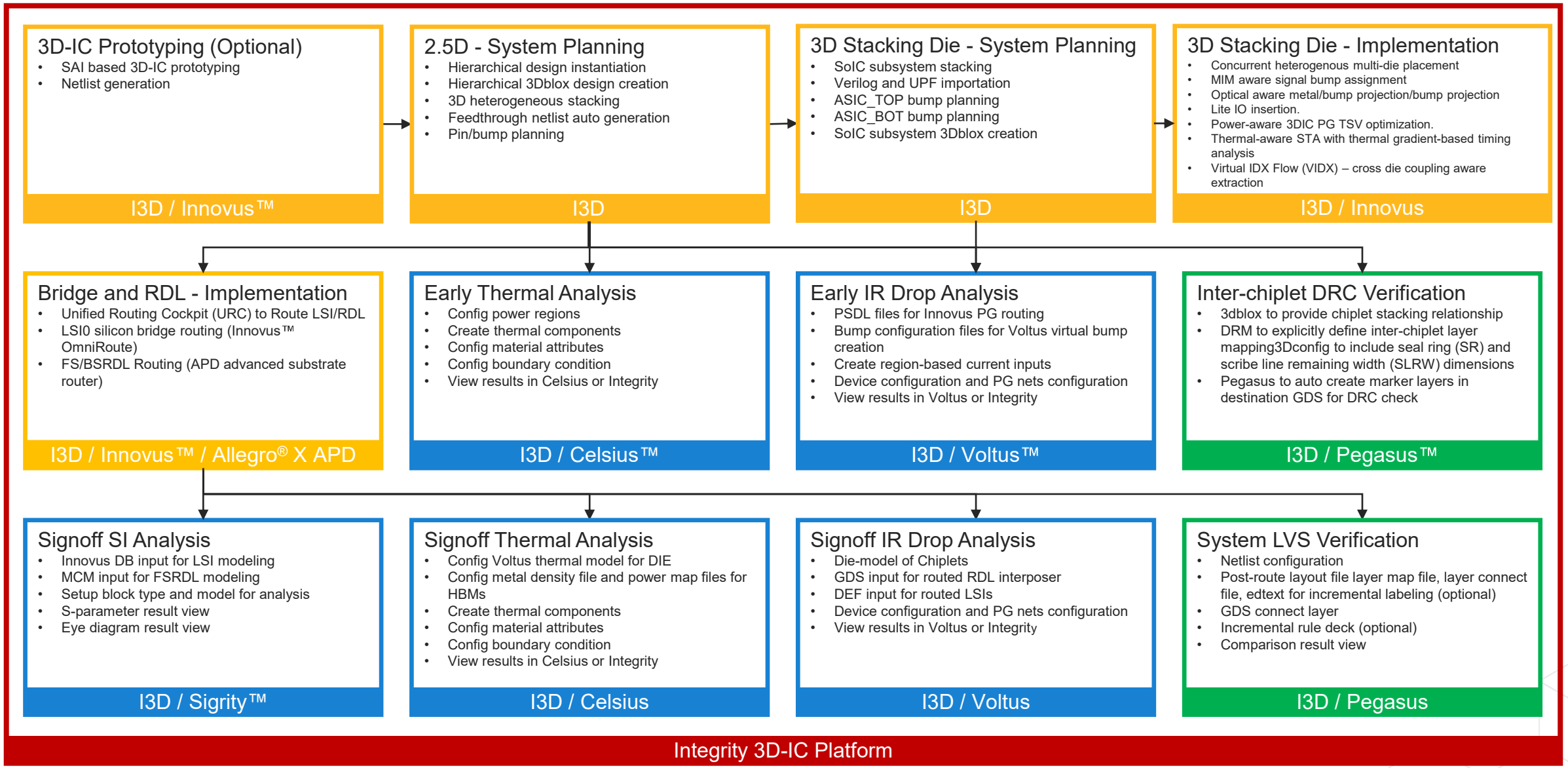




- Integrity™ 3D-IC Platform based 3.5D RAK provides a user end-to-end design methodology for a 3.5D system. Key flows/features included in the RAK:
  - 3.5D full system heterogeneous stack creation
  - System and bump planning
  - 3D stacking dies implementation, optimization, and analysis (more details to follow)
  - Unified Routing Cockpit (URC) to route both silicon bridges and RDLs
  - Early/signoff IR drop analysis
  - Early/signoff thermal analysis
  - Early/signoff system LVS
  - Inter-chiplet DRC verification
  - Signoff SI analysis
- It can serve as a generic Cadence reference flow for 3.5D stacking dies on RDL interposer and silicon bridge style process, such as TSMC CoWoS-L, Intel EMIB, Samsung I-CubeE
- 3.5D RAK early access for selected customers only

© 2026 Cadence Design Systems, Inc. All rights reserved.

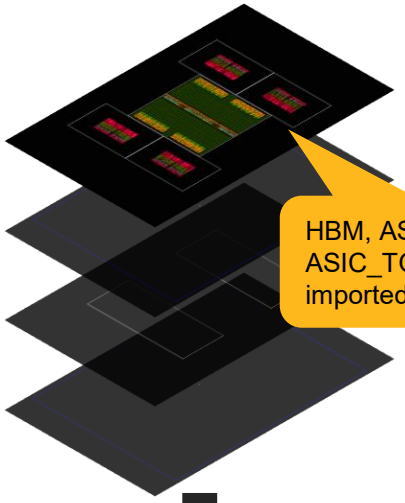
Integrity 3D-IC 3.5D RAK End-to-End Flow Chart



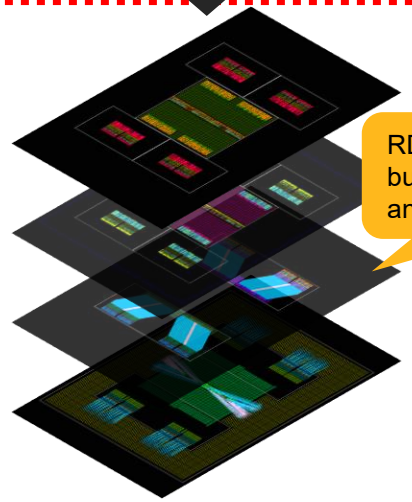
© 2026 Cadence Design Systems, Inc. All rights reserved.

Bump Planning

Pre-bump-planning design

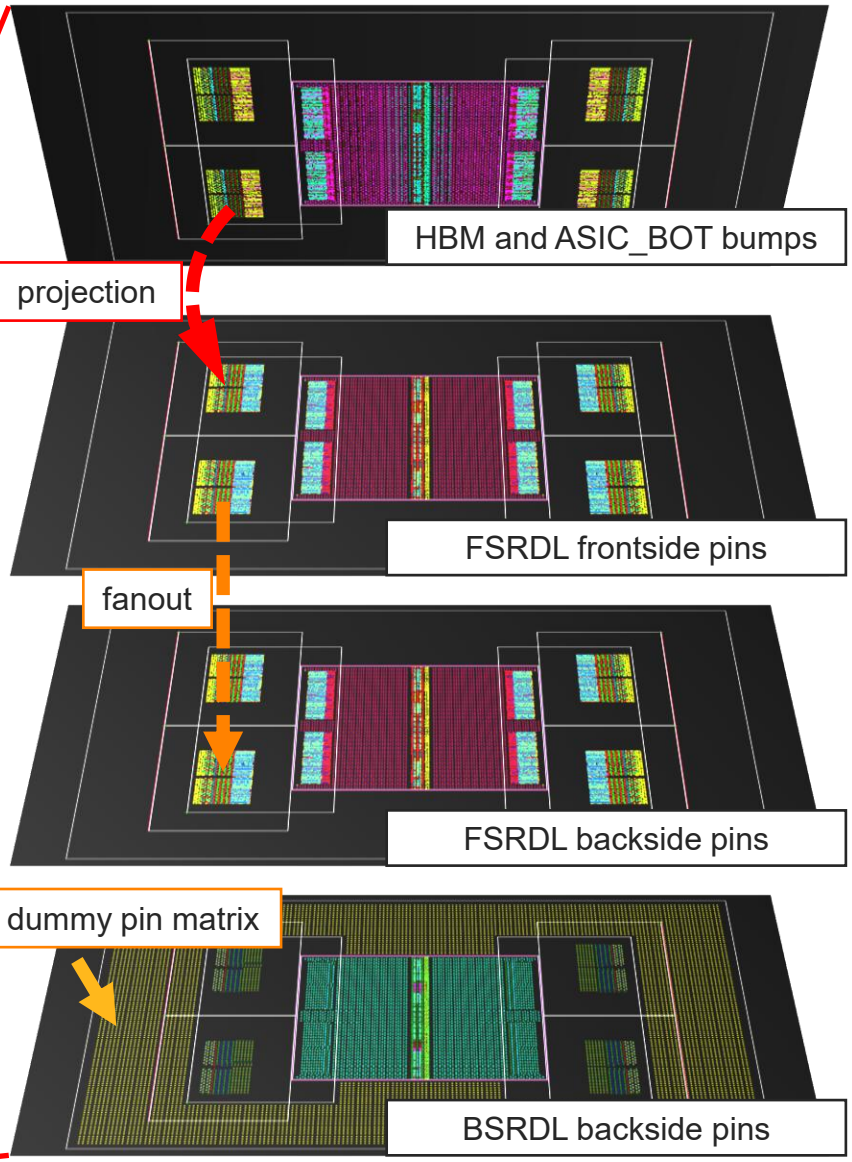


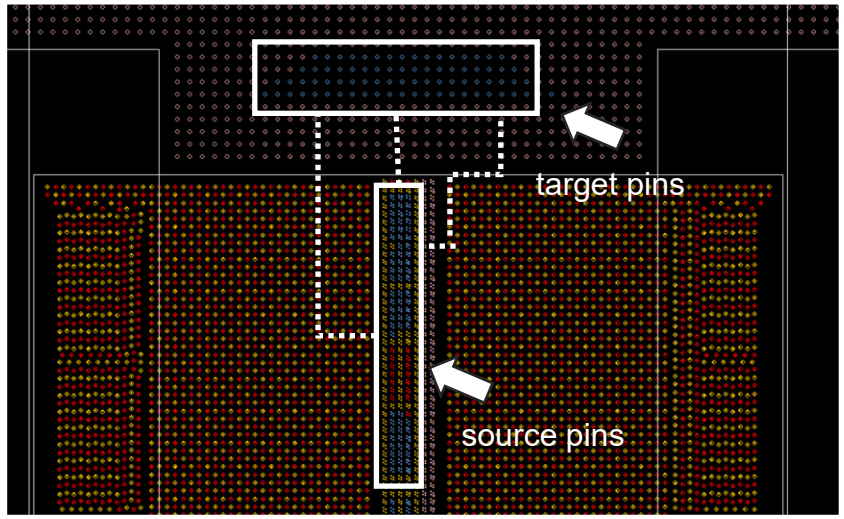
HBM, ASIC_BOT, ASIC_TOP bumps are imported.



RDL pins and LSI bumps are created and assigned.

Post-bump-planning design



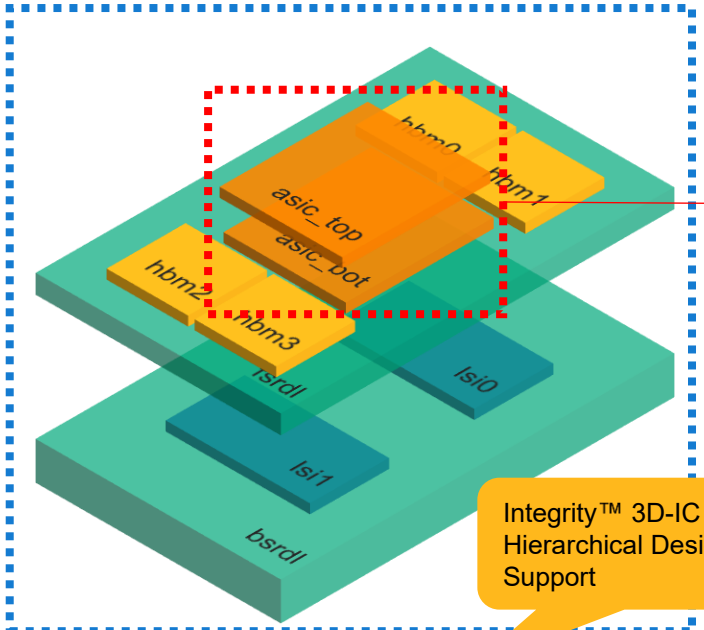


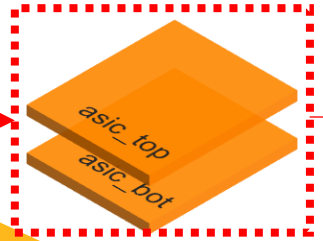
Integrity System Planner also supports bump/pin assignment optimization feature, which will automatically build connections between source and target pins to:

- avoid connection crossing
- (optional) build connections with length constraints
- (optional) do routability-aware pin assignment

© 2026 Cadence Design Systems, Inc. All rights reserved.

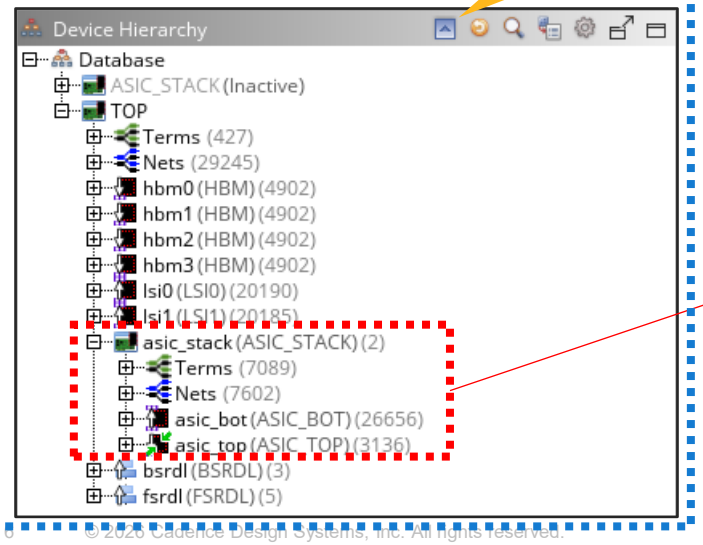
Hierarchical 3DBlox Auto Partition

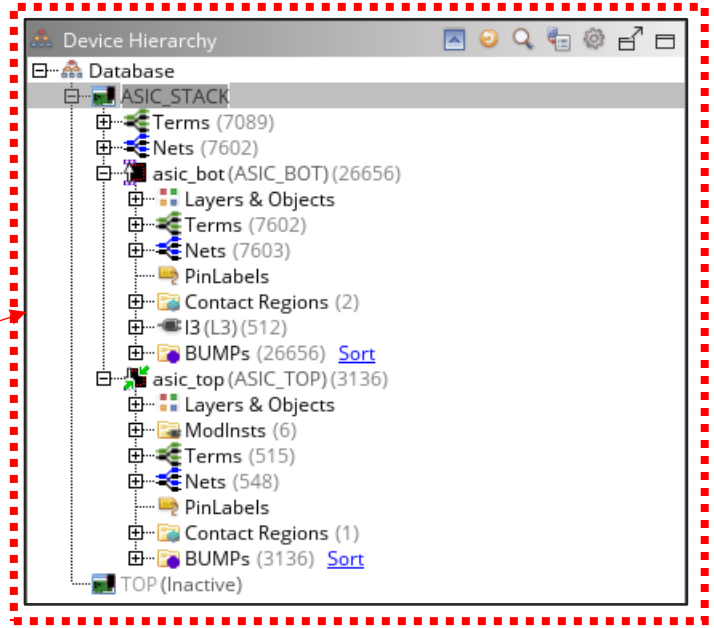




3D Stacking dies can be partitioned out as a separated and self-contained 3D-IC design

Integrity™ 3D-IC - Fully Hierarchical Design Support





Self-contained stacking dies 3D-IC design – implementation, optimization, and analysis

- Heterogeneous integration, F2F
- Bump projection
- Die-level feedthrough
- Timing budget
- UPF partitioning
- 3D placement
- Metal projection
- MIM-aware bump assignment
- Lite I/O insertion
- TSV optimization and ERA
- Thermal-aware STA
- VIDX Extraction

Assemble back to 3.5D design for full system optimization and analysis

Full 3.5D System

Stacking dies subsystem

© 2026 Cadence Design Systems, Inc. All rights reserved.

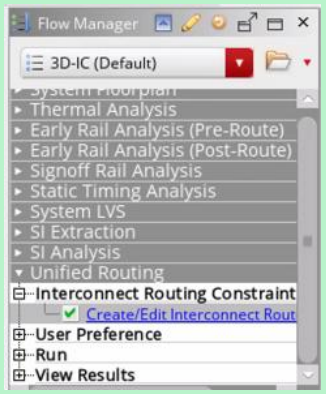
cadence

Integrity 3D-IC - Unified Routing Cockpit

Parallel routing implementation with unified routing cockpit (URC) in Integrity 3D-IC

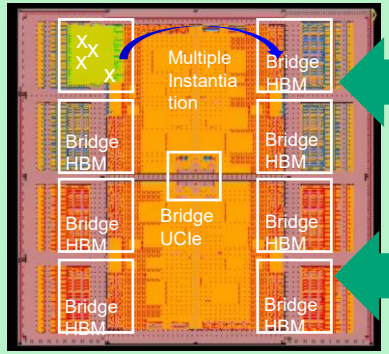
URC for Auto Routing

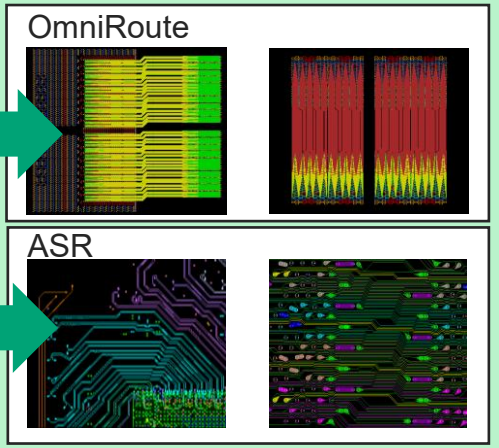
- Create/Edit Unified Constraint
- User Preference Setting
- Auto Routing
- Routed DB/DRC Back Annotation
- DRC/Connectivity Checking
- SI/PI Analysis Clarity→TopWB
- Routing Constraint Update
- Save/Restore Database



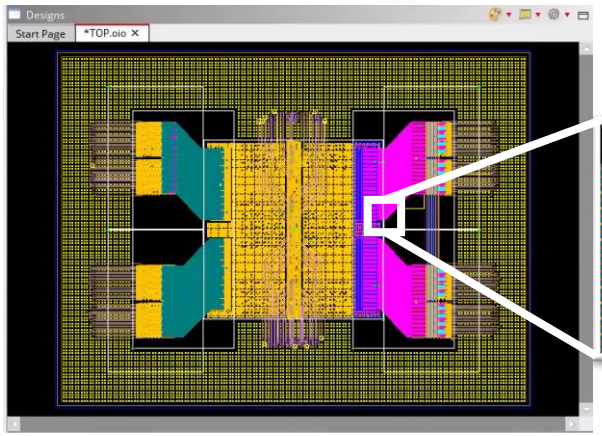
Unified Routing Cockpit

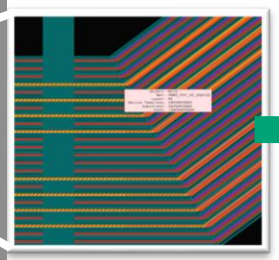
- URC
- Auto Route for Both Die/RDL
- Parallel Routing Job Control
- DRC Browsing and LVS Checking
- Fully Scriptable and Automatic Solution

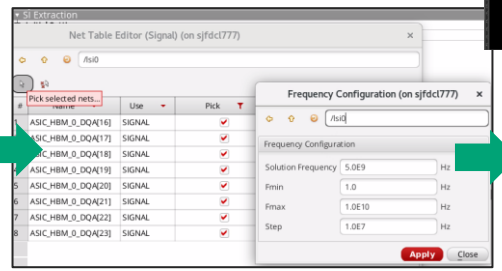


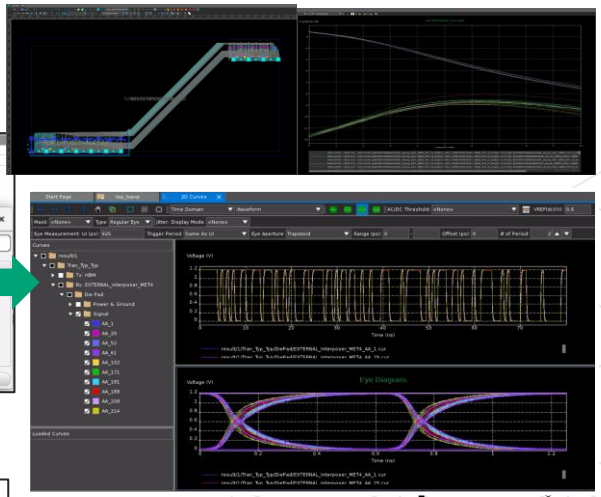


In-Design Seamless SI Analysis for Multi Net Groups in Parallel









*Required Integrity-3DIC build: **25.13-e009_1** or later version*

© 2026 Cadence Design Systems, Inc. All rights reserved.



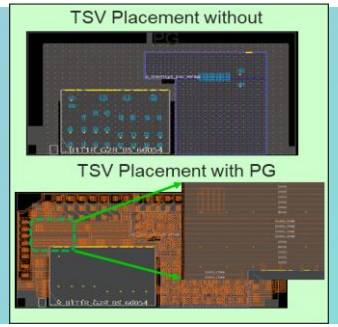
3.5D Design – Stacking-Dies Implementation, Optimization, and Analysis

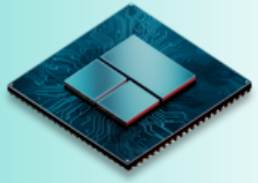
© 2025 Cadence Design Systems, Inc. All rights reserved.

3.5D RAK – Stacking Dies Key Features

3D Heterogeneous Integration and Floorplan

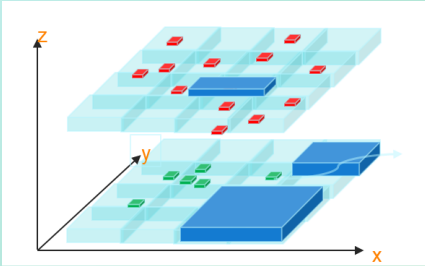
- Load and display heterogenous multiple dies
- Manage and navigate multiple devices
- PG TSV Optimization



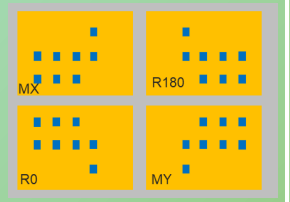


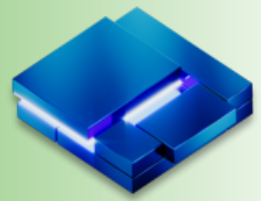
FLOORPLAN

3D Implementation



- MIM aware sig bump assignment
- Metal projection/ bump projection
- Lite IO insertion
- Timing budget and upf partitioning
- 3DIC heterogeneous concurrent multi-die placement

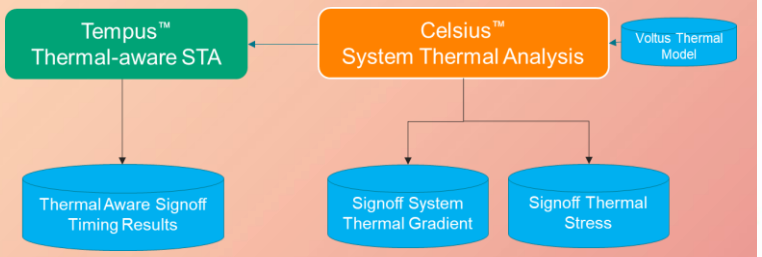




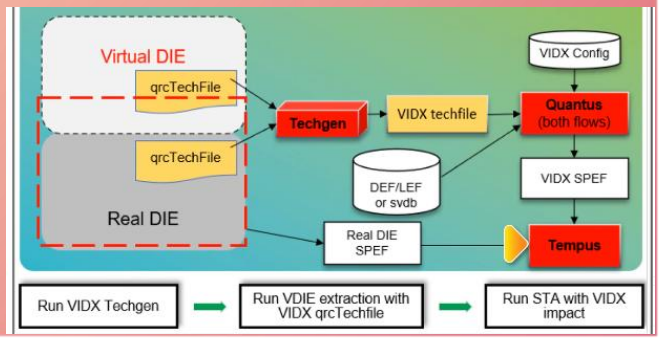
IMPLEMENTATION

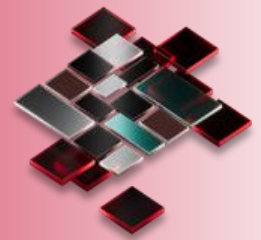
3D-IC STA

Thermal-Aware STA



Virtual IDX Flow



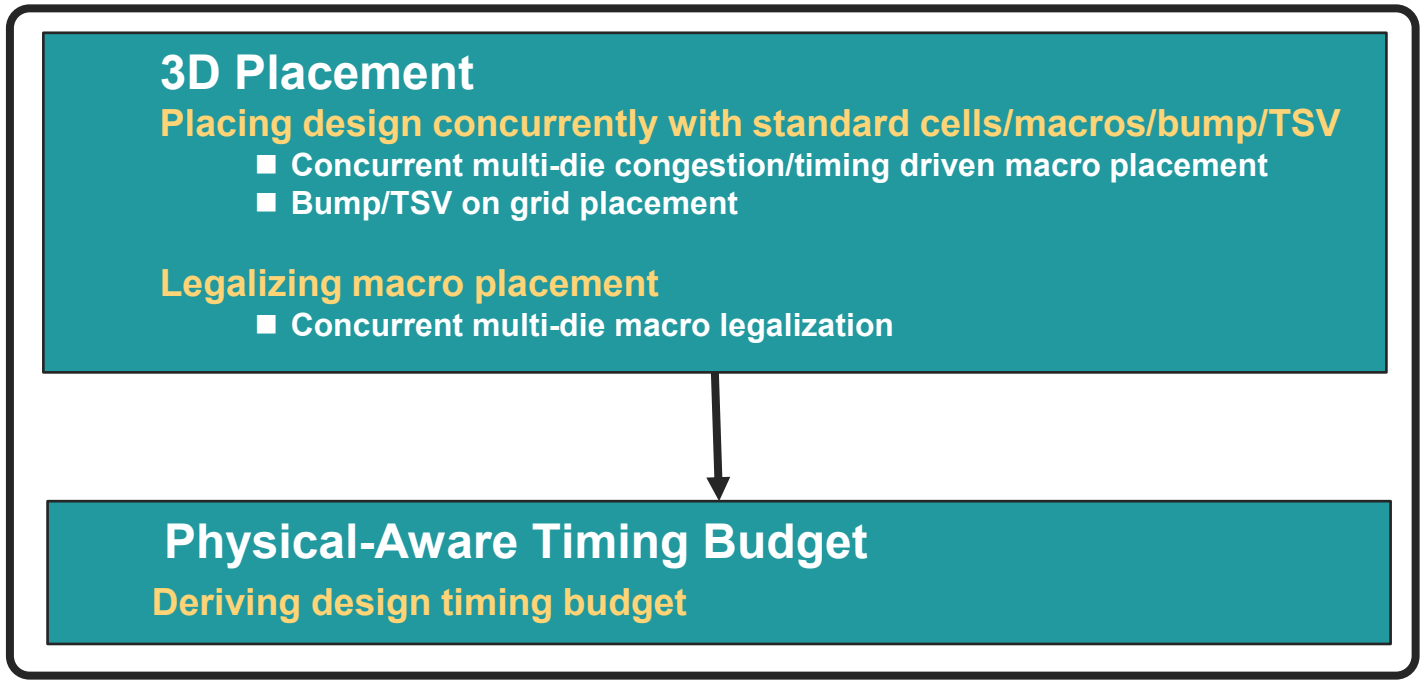


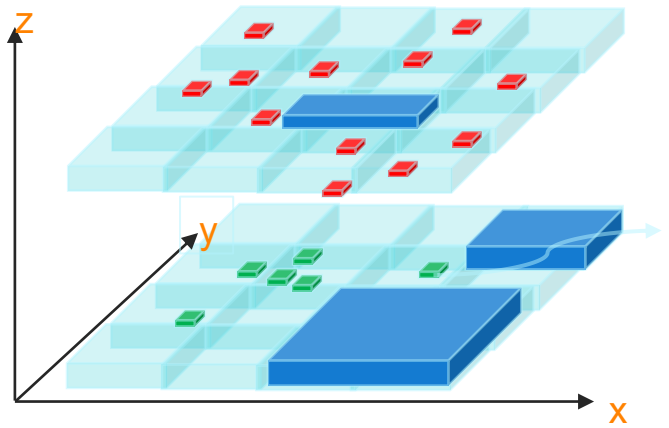
SIGNOFF

Stacking Die Solution with Integrity™ 3D-IC

© 2026 Cadence Design Systems, Inc. All rights reserved.

Stacking Dies – Concurrent Heterogenous 3D Placement Flow



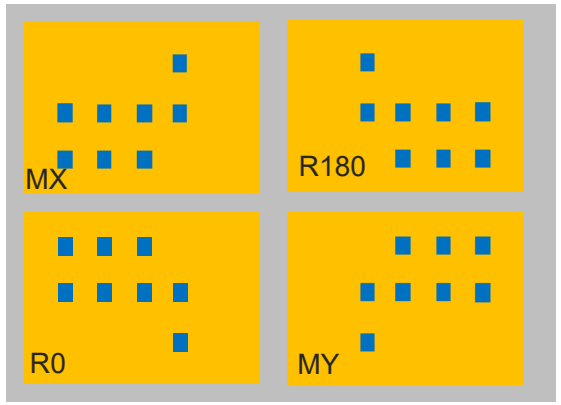


- Concurrently placing instances, macros, and bumps/TSVs across multiple dies
- Supporting the following features:
  - **Heterogenous** process nodes
  - **Timing and congestion-driven**
  - Fence, guide, region, blockage, etc.
  - **Enhanced 3D-MMMC**
  - Automatic crossing-die **feedthrough insertion**
  - Full-system **virtual timing analysis**

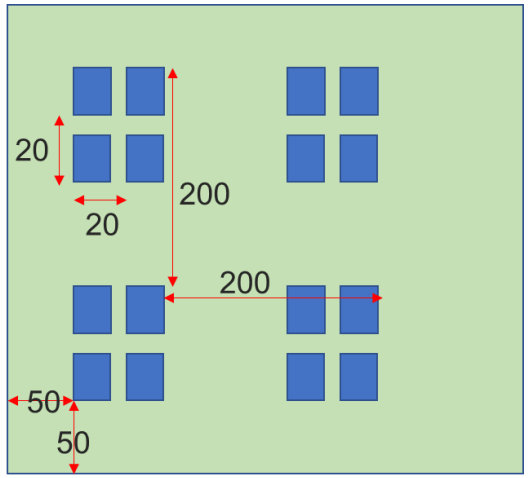
© 2026 Cadence Design S

Stacking Dies – Physical-Driven Bump Assignment

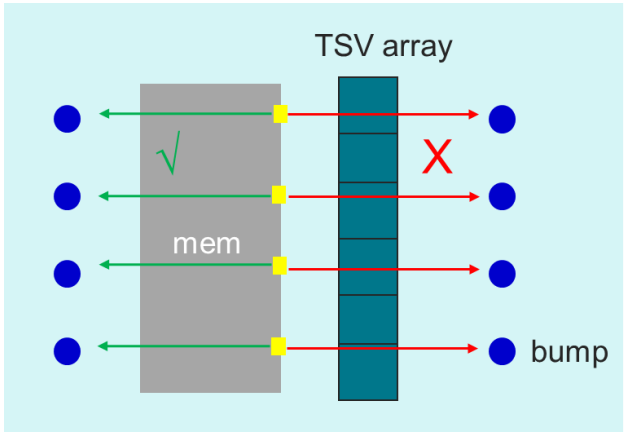
- Enabling bump assignment for heterogenous multi-die 3D-IC designs
- Including placement of blockage, PG strips, and macros configurations within the dies
- Supporting cloning/reusing of master block, bump array assignment, and routability assessment



Cloning/reusing of master block



Bump array assignment



Routability assessment

© 2026 Cadence Design Systems, Inc. All rights reserved.

Overview: TSV Distribution Optimization Function and Flow

From YAML-defined constraints to verified 3D PG TSV distribution

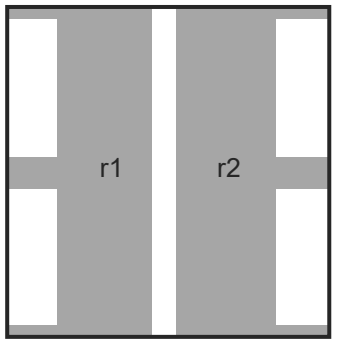
Function Overview

- Automated PG-aware TSV distribution driven by YAML Configuration
- Supports pre-PG planning and post-PG TSV placement
- Co-optimizes power demand, density constraints, and routability
- Macro-, blockage-, and power-domain-aware
- Designed for fast iteration with simple configuration
- Built-in-muti-stage check mechanism (pre / in / post distribution)

Implementation Flow Overview

Input Power and Design Data

	Power Domain	Voltage	Current
ASIC_TOP	r1	0.75	100mA
	r2	0.75	100mA
ASIC_BOT	r1	0.75	100mA
	r2	0.75	100mA

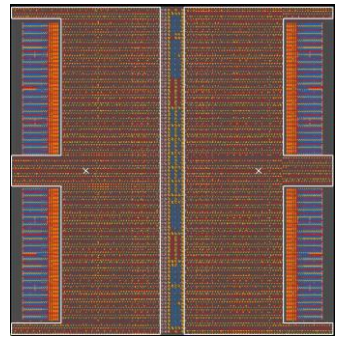


YAML-based Configuration

```
die_inst_name:
  Cell: die_cell_name
  Total_tsv_num: tsv_number
  Regions:
    region_name:
      Rects:
        -[x1, y1, x2, y2, …]
      Polygons:
        -[x1, y1, x2, y2, x3, y3, …]
  Nets:
    net_name:
      Voltage: voltage
      Bump_cell: bump_cell_name
      Regions:
        region_name:
          current_consumption: current
```

One-Command Flow

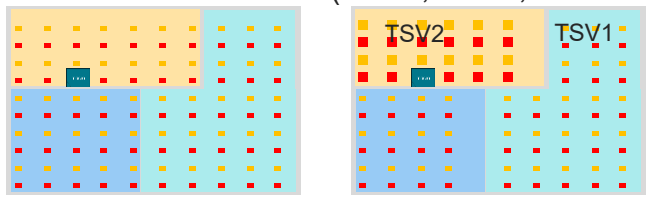
```
proto3d::opt_tsv_distribution \
  -input_file $yaml -pitch 60 \
  -pattern inline \
  -pattern_direction horizontal \
  -min_distance_between_different_pg 20 \
  -project_power_domain \
  -verbose
```



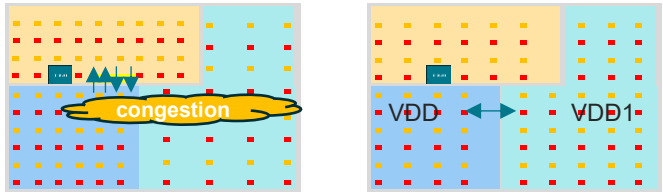
© 2026 Cadence Design Systems, Inc. All rights reserved.

End-to-End 3D PG TSV Distribution Optimization

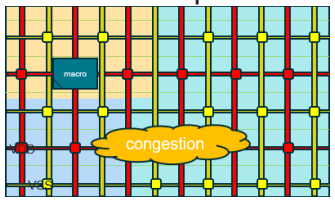
Pre-PG TSV distribution (baseline)
- Region-based power distribution (ex. 3 regions)
- Bump grid-aligned TSV placement
- horizontal (or vertical) direction (in-line or stagger)
- Macro-aware and blockage-honored placement
- Mixed TSV dimensions (TSV1, TSV2, TSV3…)



Congestion-aware TSV distribution optimization
- Local density constrained (YAML-defined)
- Avoid partition pin regions
- Routability-aware PG alignment (ex: 1P1G)
- Power-domain-aware considering different power

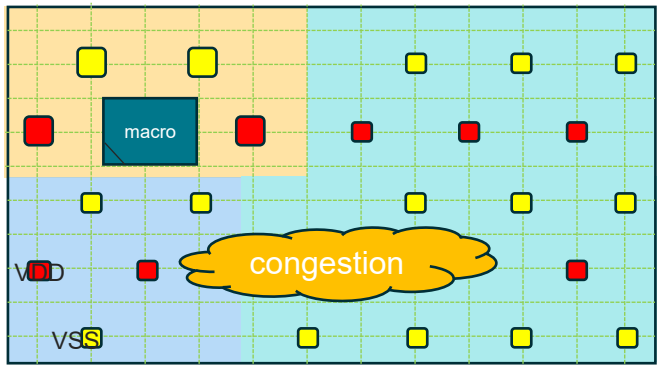


Power-Plan-aware TSV distribution optimization
- Post-PG TSV placement (landing-layer aware)



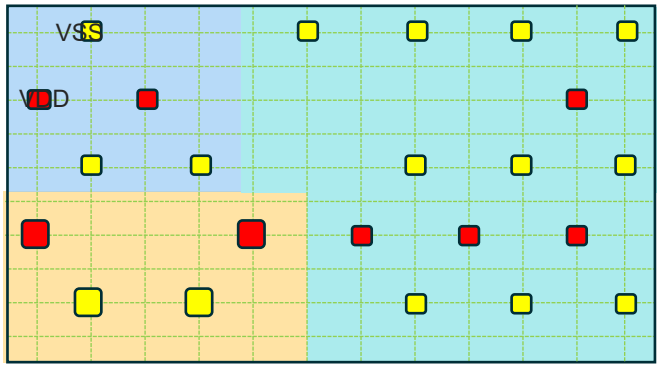
3D PG TSV Distribution Optimization
- Top-die power domains projected to bottom die
- Top-die PD shrunk by 80% for projection mapping
- Congestion-aware under direct projection

Power Domain Projection from Top-Die



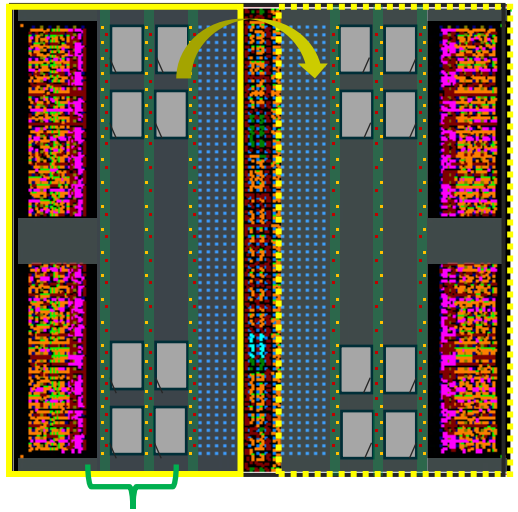
TOP-Die Shrink 80%

Top-Die PD projects to Bottom-Die



Advanced applications
- Identical Channel Pattern and Symmetric Cloning

Clone to right – Symmetric layout



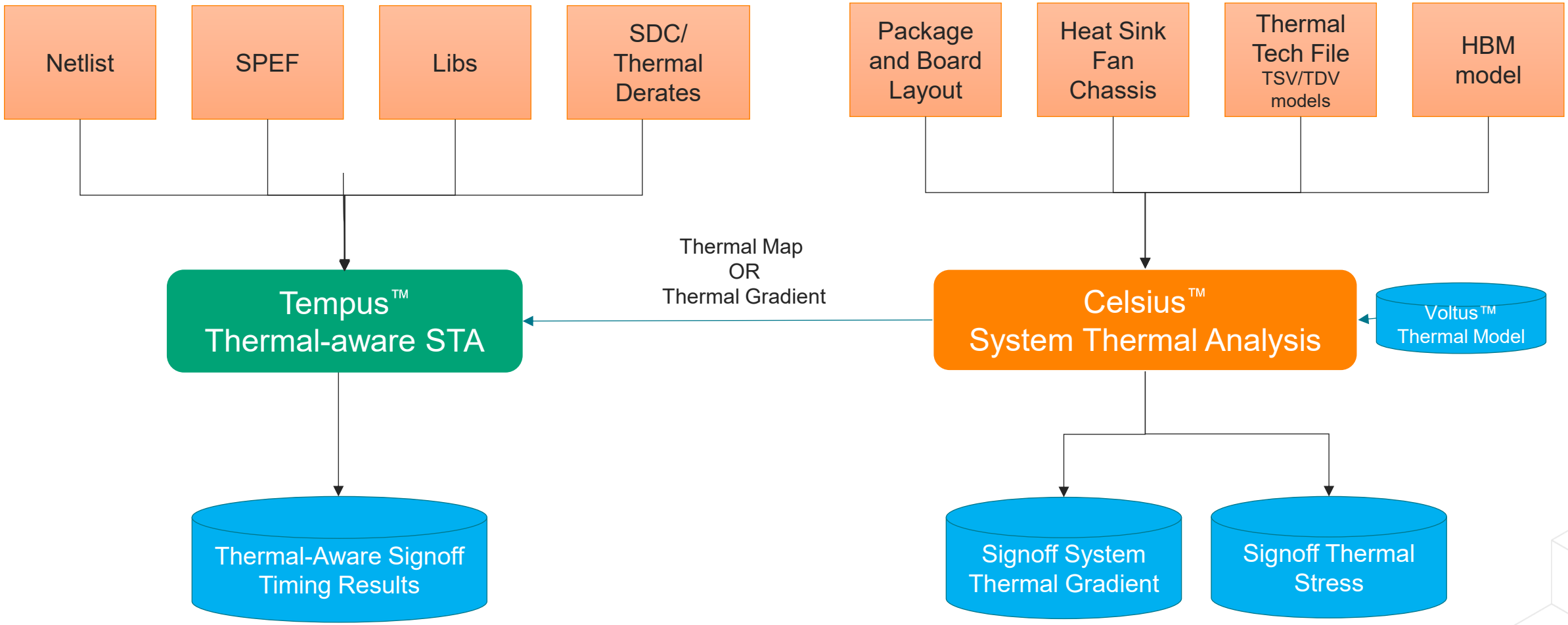
Left region - Identical channel pattern

TSV density constraints and check result
- TSV local density verification report

```
############################################################
# die_cell bottom_die - (allowable local TSV density - upper bound) \
    = (original TSV local_max density constraint) - (estimated signal TSV density)
# die_cell bottom_die - (allowable local TSV density - lower bound) \
    = (original TSV local_min density constraint) - (estimated signal TSV density)
# die_cell bottom_die - estimated signal TSV density : 0.0096875
# die_cell bottom_die - original TSV local_max density constraint : 0.05
# die_cell bottom_die - allowable local TSV density - upper bound : 0.0403125
# die_cell bottom_die - TSV window width : 108um
# die_cell bottom_die - TSV window step : 54um
############################################################
die_cell        density_type permissible_range         window_density  window_box
--------------- ------------ ------------------------- --------------- -------------------
--------------
bottom_die      tsv          0-0.0403125               0.0214814814815 {270 108 320 216}
bottom_die      tsv          0-0.0403125               0.0207407407407 {270 54 320 162}
bottom_die      tsv          0-0.0403125               0.0207407407407 {270 162 320 270}
bottom_die      tsv          0-0.0403125               0.02            {270 216 320 320}
bottom_die      tsv          0-0.0403125               0.0192          {270 270 320 320}
```

cadence

Thermal-Aware STA for 3D-IC Signoff



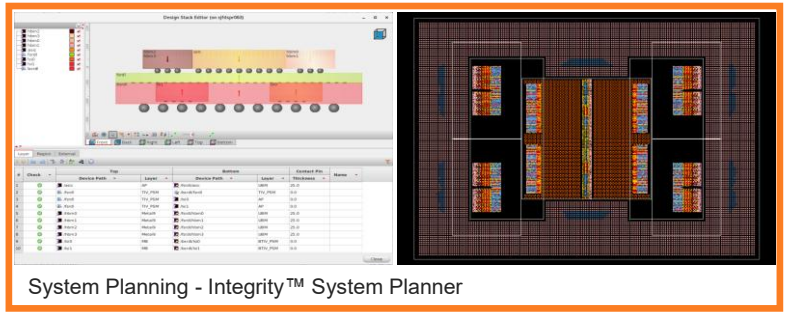
© 2026 Cadence Design Systems, Inc. All rights reserved.



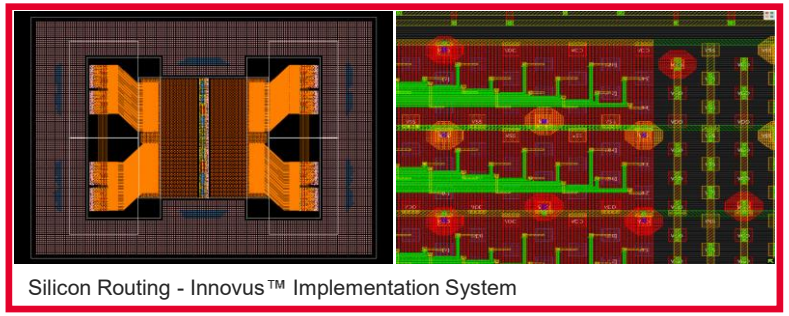
3.5D Design – Full System Analysis and Verification

© 2025 Cadence Design Systems, Inc. All rights reserved.

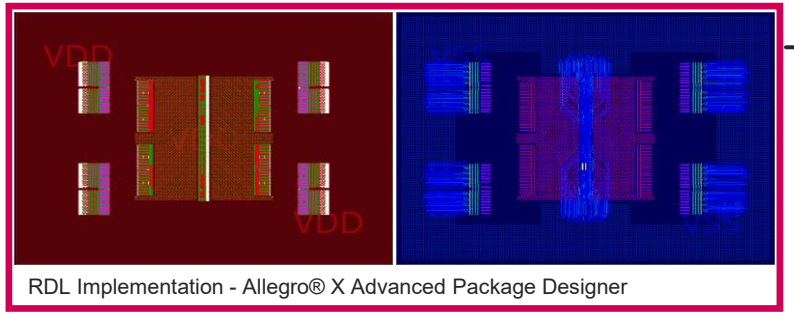
3.5D RAK – Full System Analysis and Verification



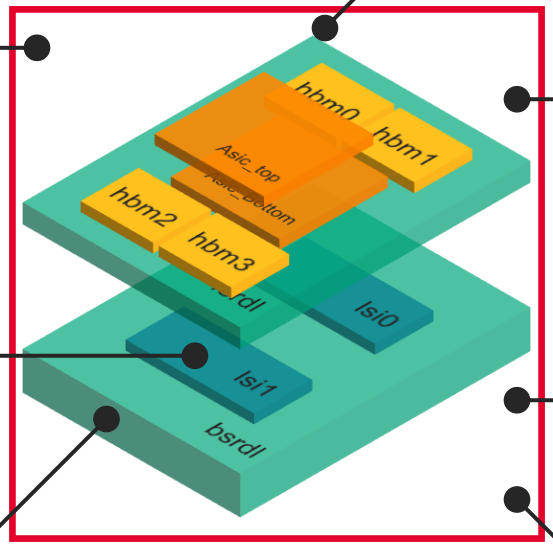
System Planning - Integrity™ System Planner



Silicon Routing - Innovus™ Implementation System

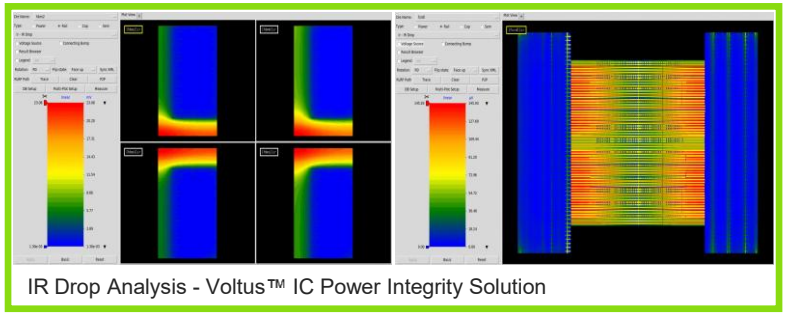


RDL Implementation - Allegro® X Advanced Package Designer

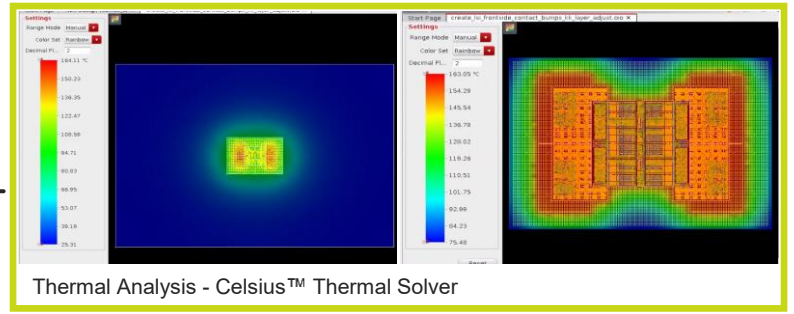


Integrity™ 3D-IC Platform

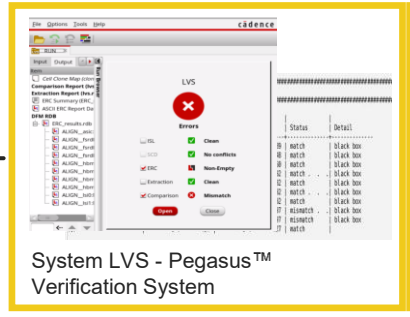




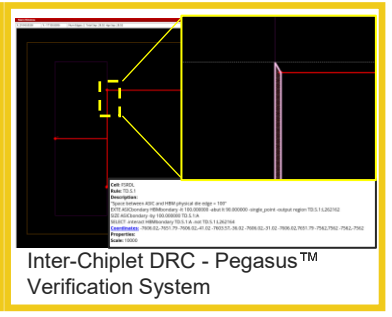
IR Drop Analysis - Voltus™ IC Power Integrity Solution



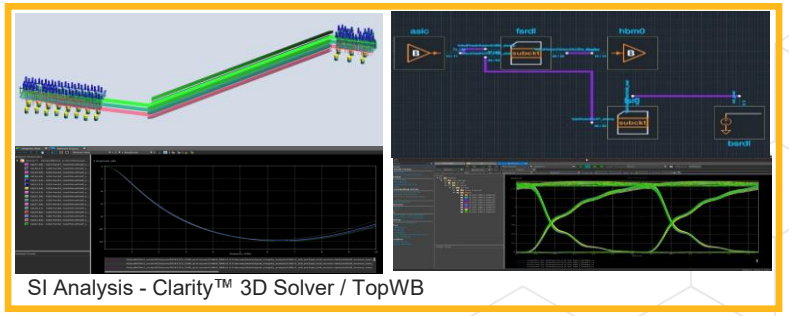
Thermal Analysis - Celsius™ Thermal Solver



System LVS - Pegasus™ Verification System



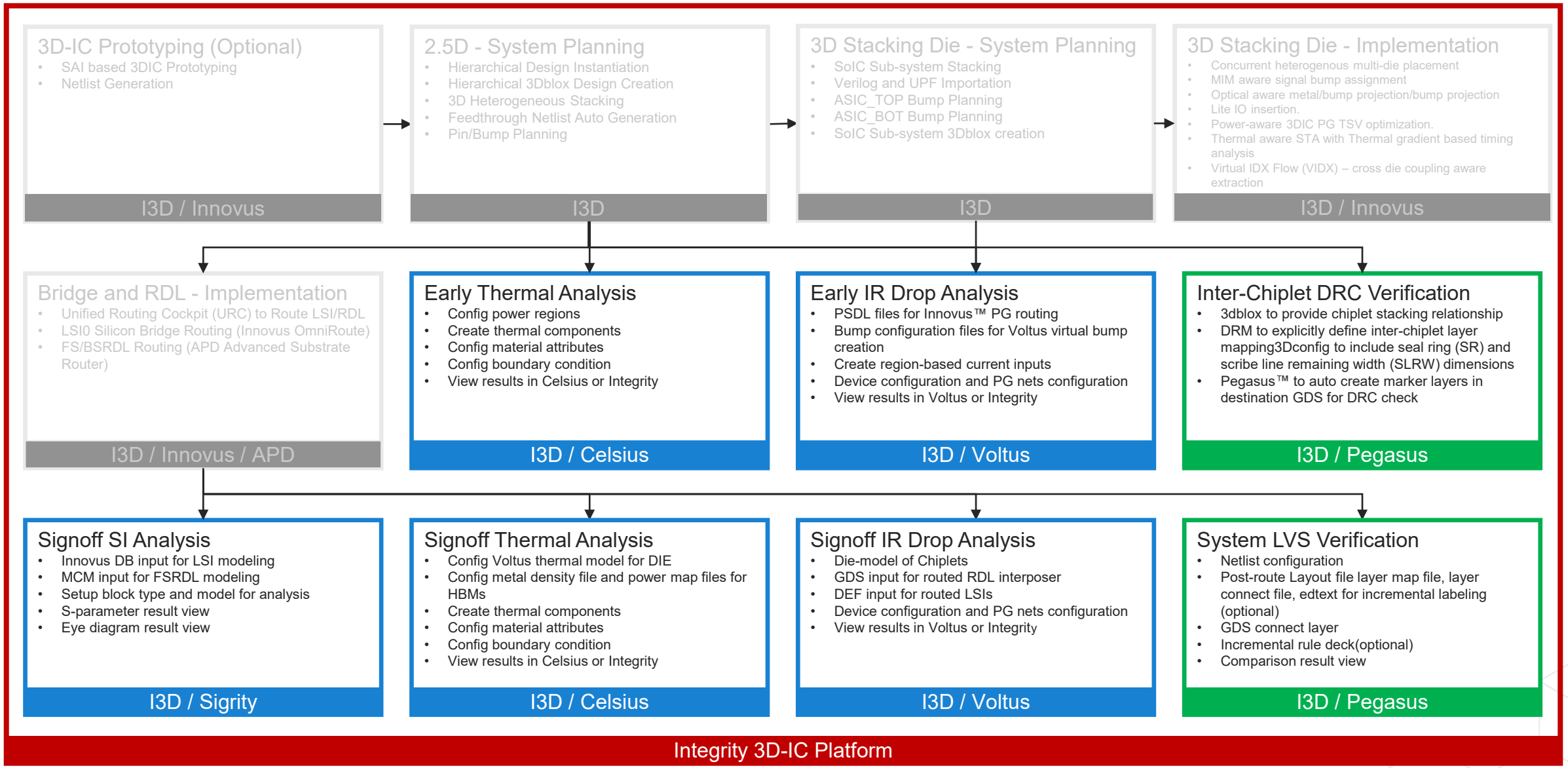
Inter-Chiplet DRC - Pegasus™ Verification System



SI Analysis - Clarity™ 3D Solver / TopWB

© 2026 Cadence Design Systems, Inc. All rights reserved.

Integrity 3D-IC 3.5D RAK End-to-End Flow Chart



© 2026 Cadence Design Systems, Inc. All rights reserved.

Early IR Drop Analysis

iHDB tag: system_planning

Load 3D-IC design from iHDB

Voltage source configuration

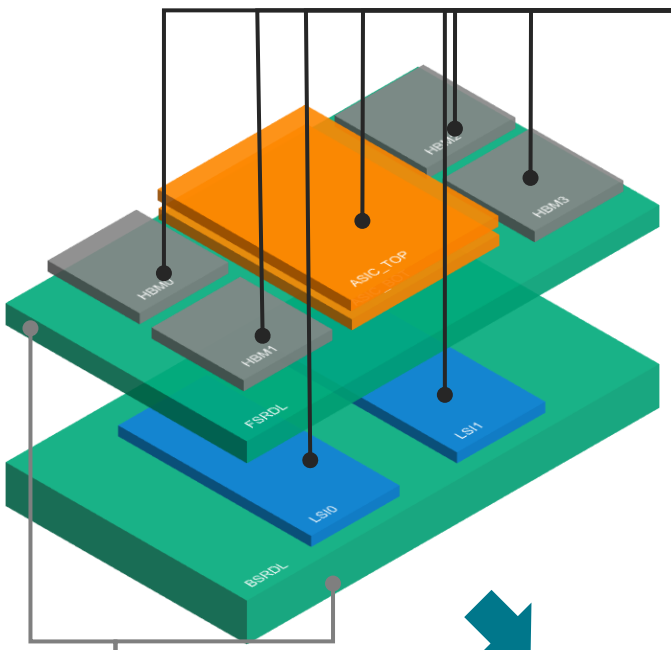
Die-level configuration
- QRC techfile
- General layer map file config
- PGV library
- P/G net configuration
- RDL GDS data

Create current region

Analysis configuration

Run analysis

View or load analysis results



GDS

Support GDS direct input for RDL design

PSDL for quick early PG routing implementation

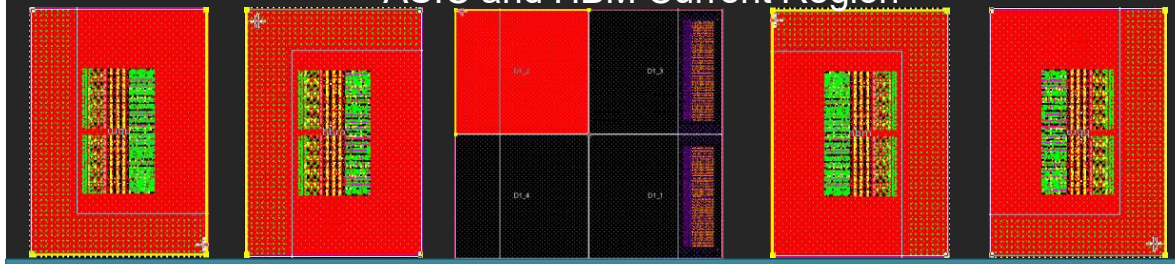
Innovus™

PSDL

PG Network Synthesis (route_pg)

ASIC, HBM, LSI DEF Files

ASIC and HBM Current Region



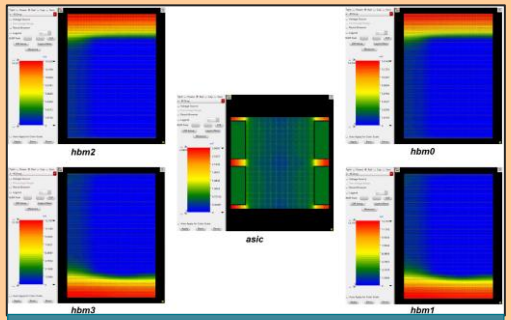
Flexible Region-based current as the power input

Die-Die mapping files

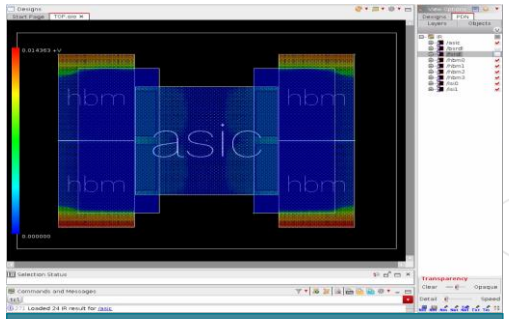
PLOC files

I3D Auto Generated

Voltus™



IR-drop analysis results – Voltus DGUI



Results back–annotation in I3D canvas

© 2026 Cadence Design Systems, Inc. All rights reserved.

cadence

Signoff IR Drop Analysis

iHDB tag: implementation

Load 3D-IC design from iHDB

Voltage source configuration

Device configuration
- QRC techfile
- General layer map file config
- Silicon die DEF/ die model
- PGV library
- RDL GDS data

P/G net configuration

Set power data

Analysis configuration

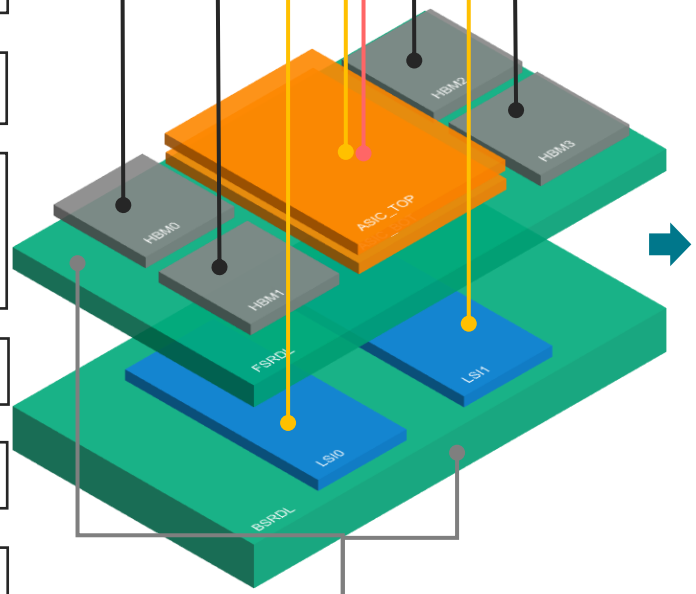
Run analysis

View or load analysis results

DEF

Power Data (static_*.ptiavg) Instance Power File (ipf)

Die Model



Die-Die mapping files

PLOC files

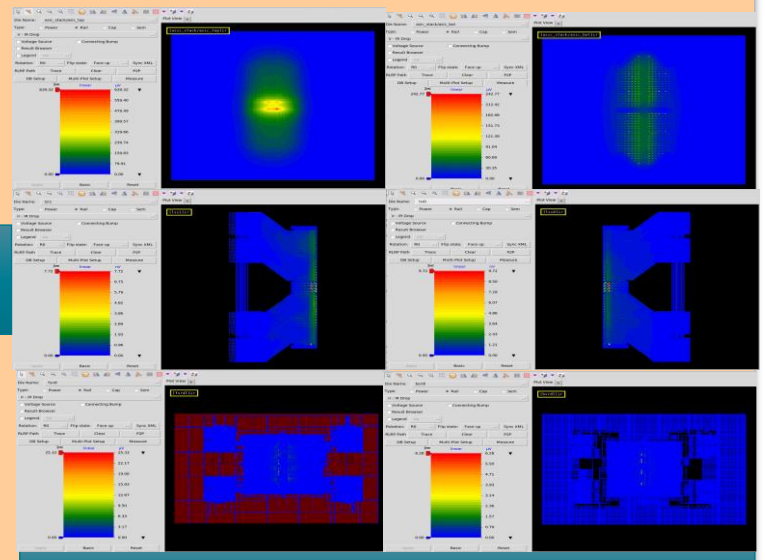
I3D Auto Generated

GDS

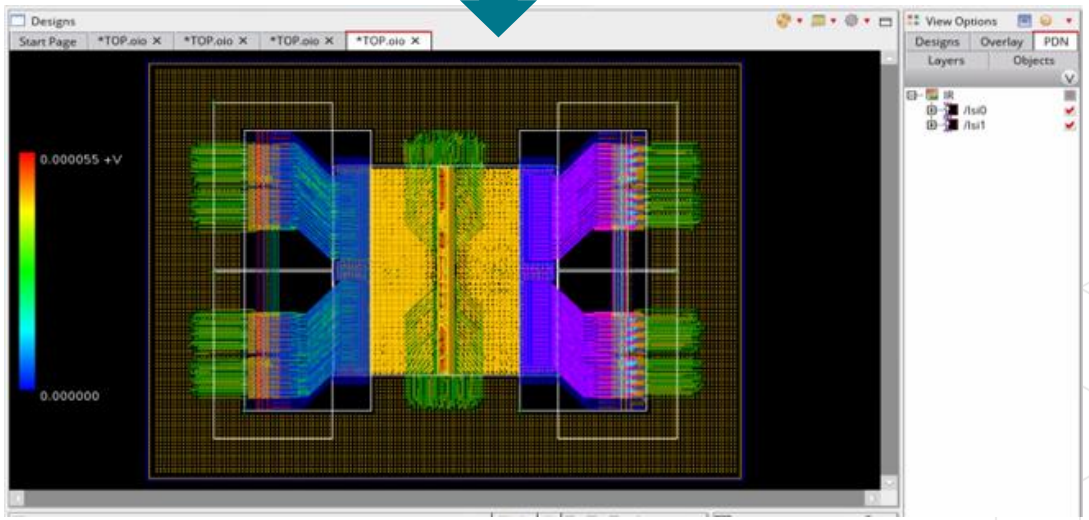
Support GDS direct input for RDL design

Voltus™

Static / Dynamic IR-drop Anaysis



IR-drop analysis results – Voltus DGUI



Results back–annotation in I3D canvas

© 2026 Cadence Design Systems, Inc. All rights reserved.

cadence

Early Thermal Analysis

iHDB tag: system_planning

Read system planning iHDB

Config analysis
- QRC techfile and general layer map file
- Power definition
- Boundary conditions
- Material assignment

Create thermal components

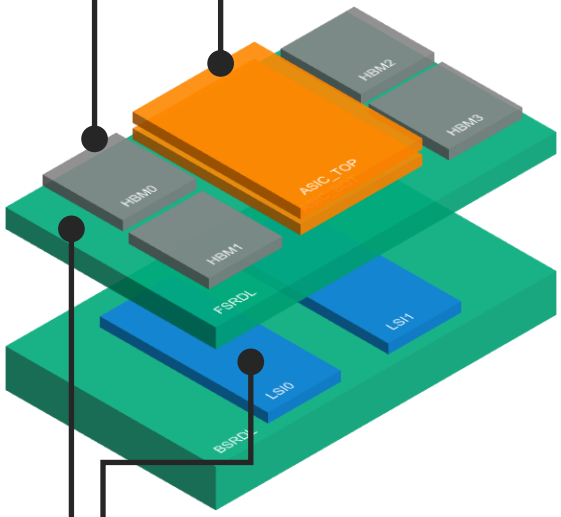
Run analysis

View or load analysis results

HBM: Region-based power

ASIC: Region-based power

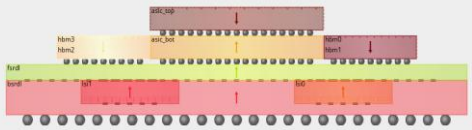
ASIC: QRC tech file for stack-up



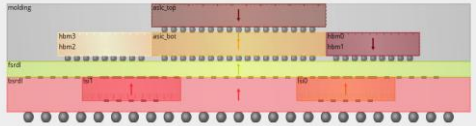
Contact layers: config bump and underfill material

RDL/LSI: QRC tech file for stack-up

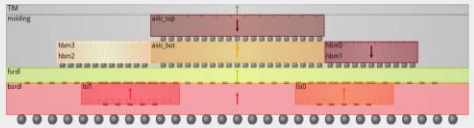
Thermal Components



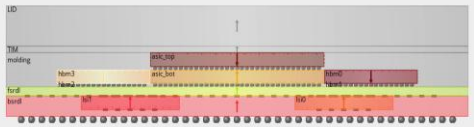
Original design after system planning



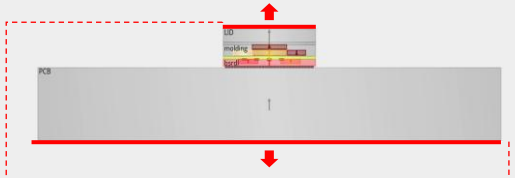
Molding compound created



TIM created



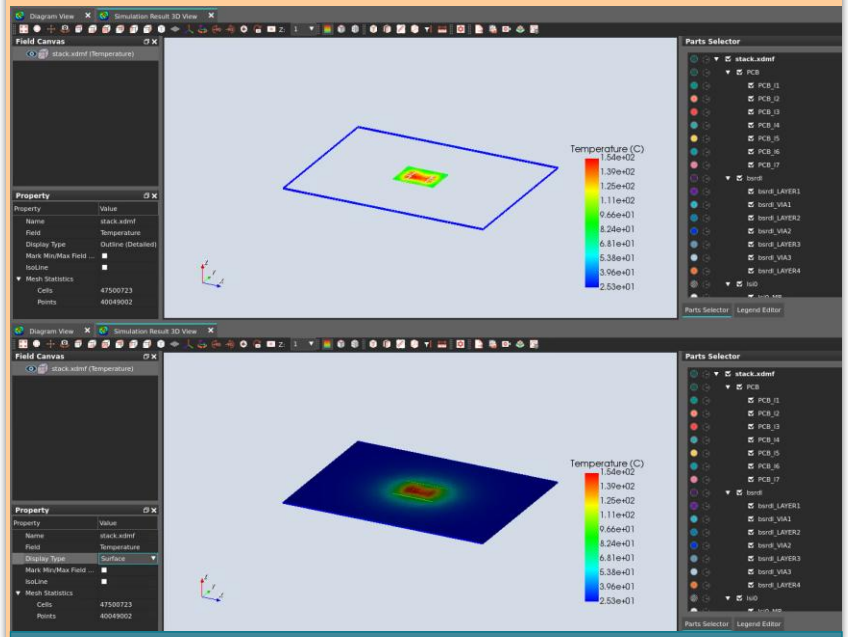
LID created



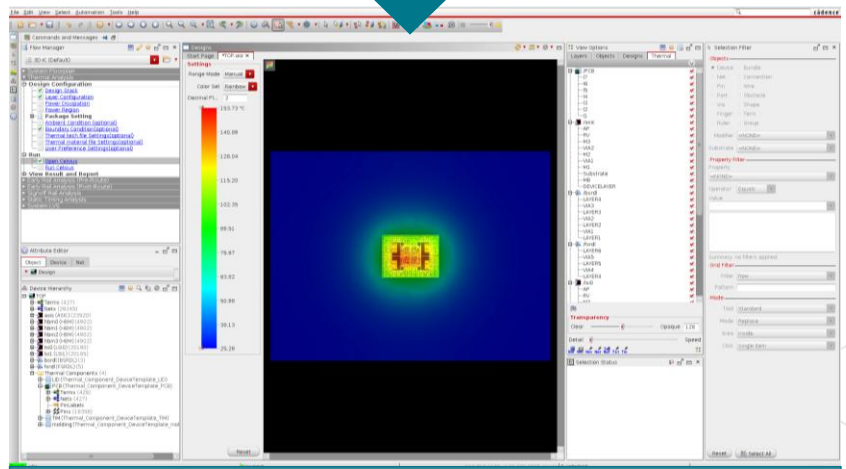
Virtual PCB created

Boundary conditions

Celsius™



Thermal analysis results: 3D view in Celsius



Thermal analysis results: back-annotation in I3D

© 2026 Cadence Design Systems, Inc. All rights reserved.

Signoff Thermal Analysis

iHDB tag: implementation

Read implementation iHDB

Create HBM slice cells

Config analysis
- VTM
- Power map file/ metal density file
- Boundary conditions
- Material assignment

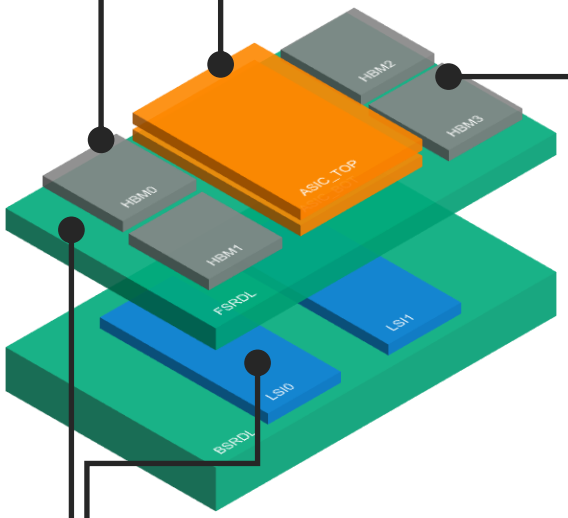
Create thermal components

Run analysis

View or load analysis results

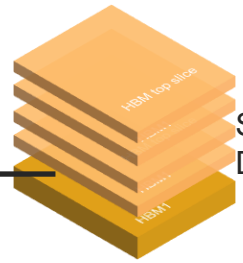
HBM: power map file and Metal density file / VTM

ASIC: VTM (Voltus Thermal Model)



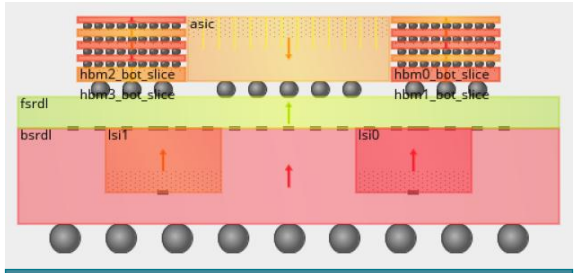
Contact layers: config bump and underfill material

RDL/LSI: mental density file

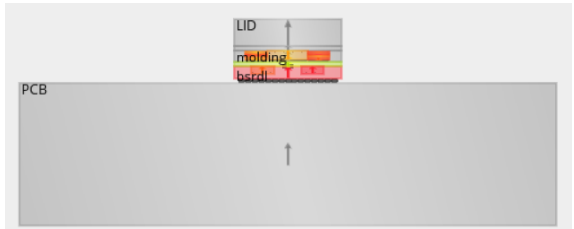


Stacked Analog dies
Digital Base die

More detailed HBM Stacking in I3D

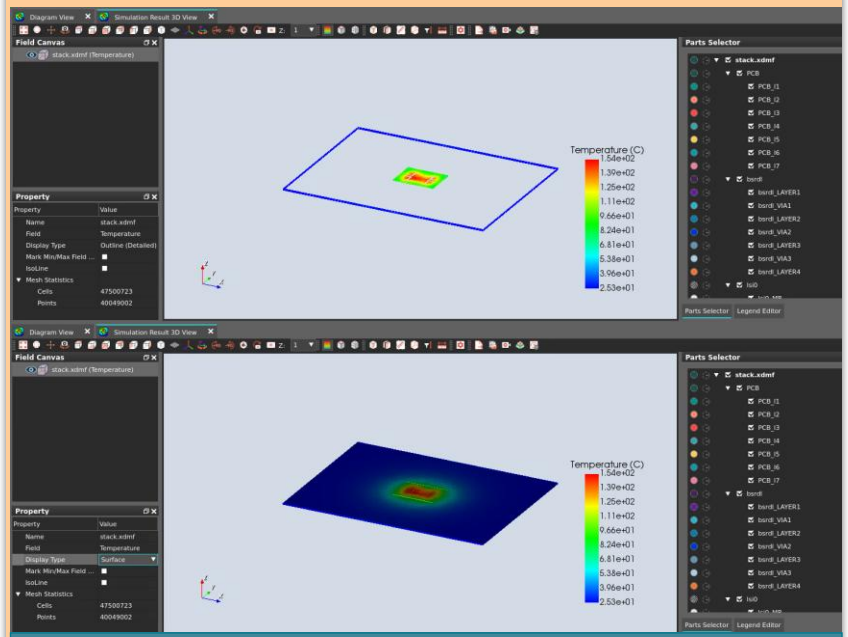


Design with multi-stacked dies as HBM

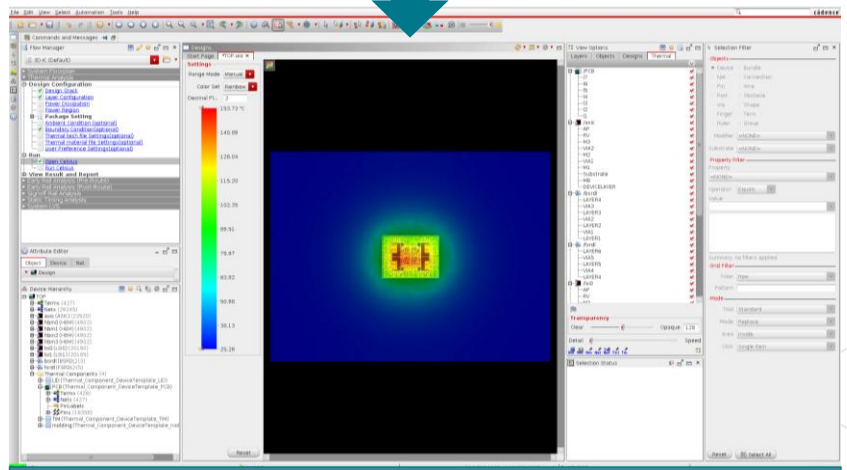


Design with thermal components created

Celsius™



Thermal analysis results: 3D view in Celsius



Thermal analysis results: back-annotation in I3D

© 2026 Cadence Design Systems, Inc. All rights reserved.

Signoff SI Analysis

iHDB tag: implementation

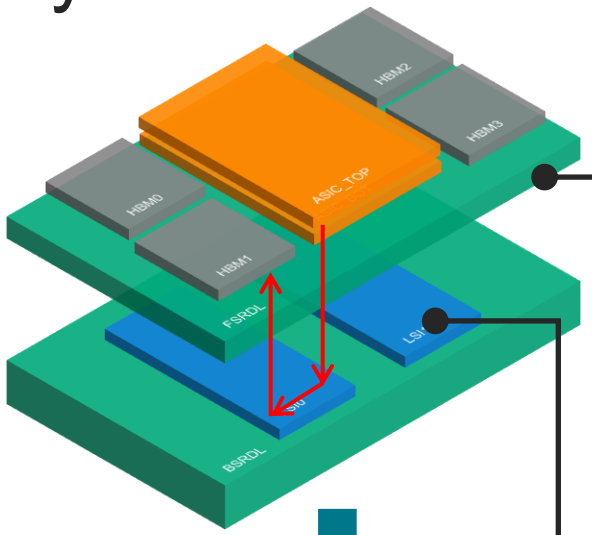
Load 3D-IC design

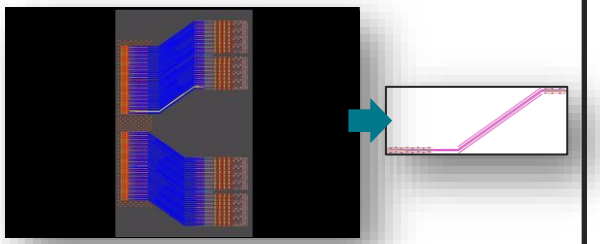
LSI0 S-parameter modeling configuration
- ICT file
- Pnr-type module model
- Signal nets and ground nets
- Layout cut margin
- Frequency settings

FSRDL S-parameter modeling configuration
- Pkg-type module model(.mcm)
- Signal nets and ground nets
- Layout cut margin
- Frequency settings

Time domain analysis configuration
- Model file for each topology block
- Analysis options YAML config file
- Eye diagram aperture width threshold

Run analysis



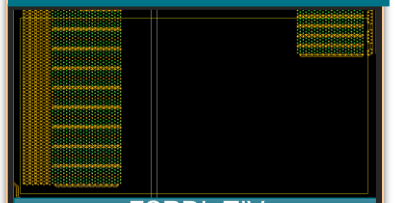


Direct-GDS Cut and Auto Stream-Out with single General Layer Map

```
oio_layer,ict_layer,gds_layer,gds_laye
r_type,gds_text_layer,gds_text_type
AP,m4,80,0,90,0
RV,v3,71,0
M3,m3,59,0,69,0
VIA2,v2,50,0
M2,m2,38,0,48,0
VIA1,v1,29,0
M1,m1,17,0,27,0
TSV,tsv,251,0
MB,mb,3,0,13,0
```

Clarity™ 3D Solver

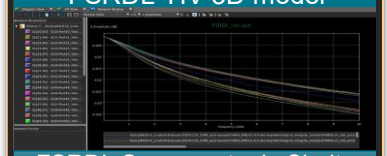
S-parameter Model Extraction



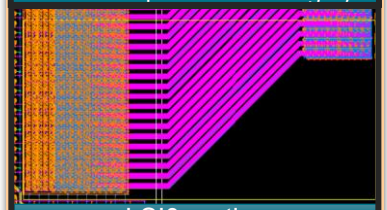
FSRDL TIV



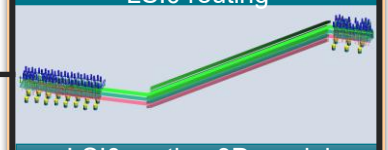
FSRDL TIV 3D model



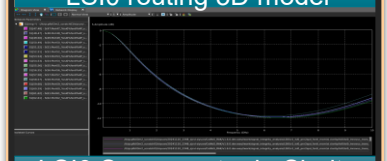
FSRDL S-parameter in Sigrity



LSI0 routing



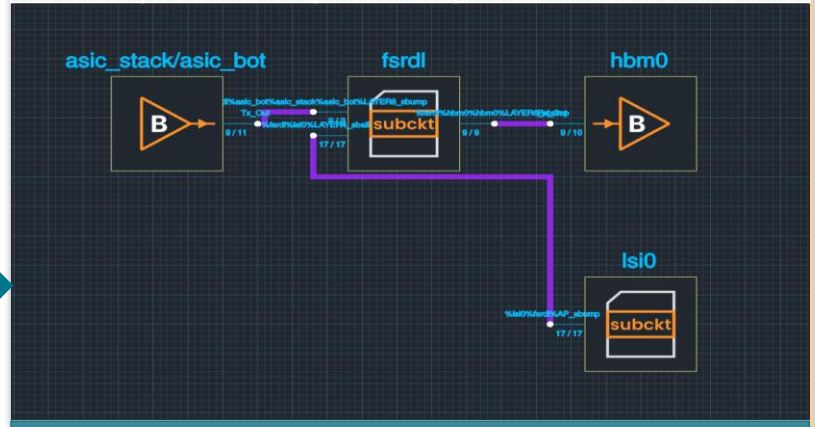
LSI0 routing 3D model



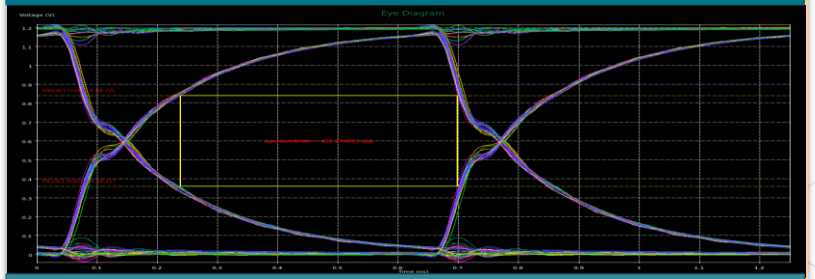
LSI0 S-parameter in Sigrity

Sigrity™ Topology Workbench

Device	Extraction Tool	Block Type	Required Collaterals
ASIC	-	Tx_ibis	IBIS file
HBM0	-	Rx_ibis	IBIS file
FSRDL	Clarity 3D Layout	Ckt	MCM (saved as pkg type module model in iHDB)
LSI0	Clarity 3D Layout	Ckt	Innovus™ DB (saved as pnr type module model in iHDB)



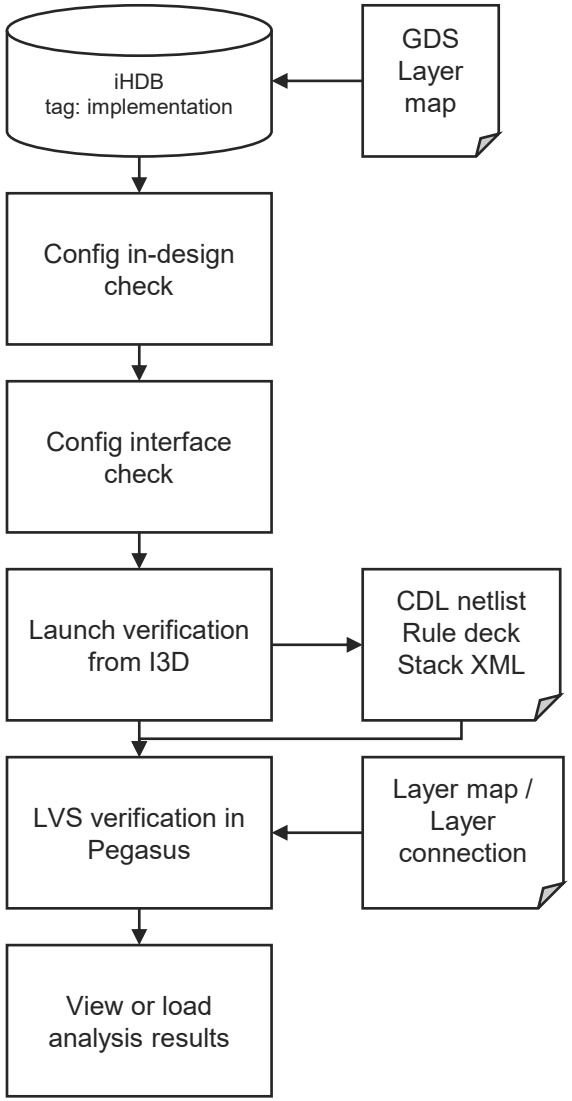
Signal path topology

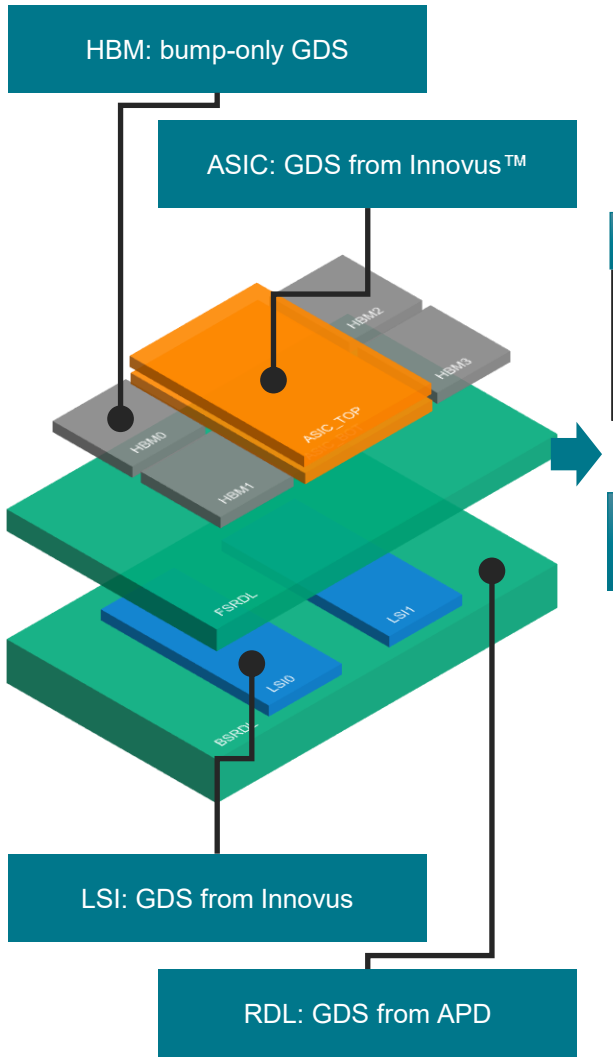


Time-domain SI analysis result eye diagram

© 2026 Cadence Design Systems, Inc. All rights reserved.

Signoff System LVS





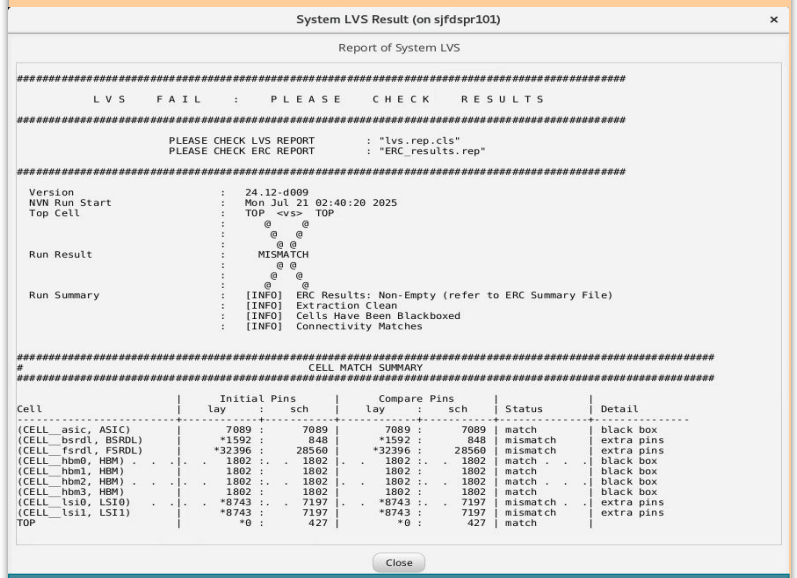
Rule-deck Free Flow

Pegasus System LVS rule deck automatically generated

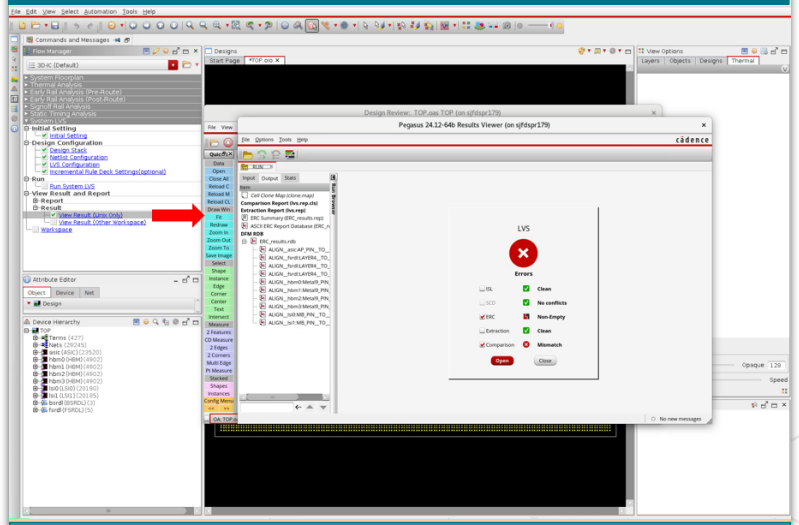
System Netlist auto-gen

CDL netlists

Pegasus™ System



System LVS Text Report



System LVS Result Viewer

© 2026 Cadence Design Systems, Inc. All rights reserved.

Signoff Inter-Chiplet DRC

3Dblox input with DRC rule deck

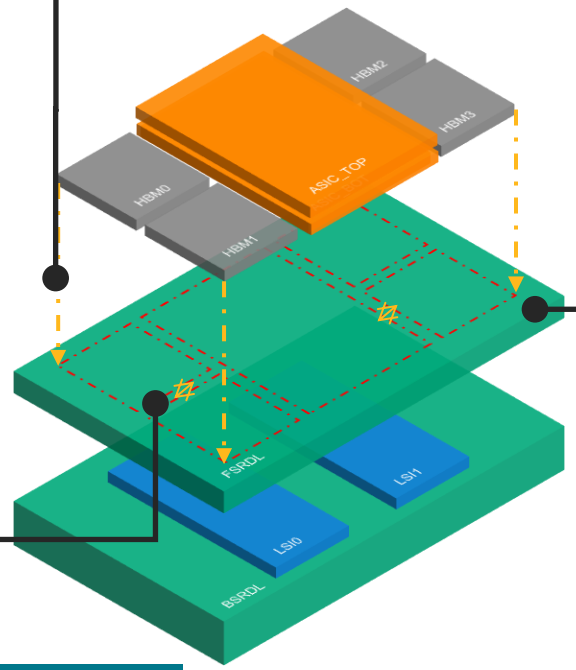
iHDB
tag: implementation

Side file for marker layer projection

Inter-chiplet DRC verification in Pegasus™

Viewing verification results

Project cross-chiplet marker layer shapes into the destination design layout automatically



Design GDS

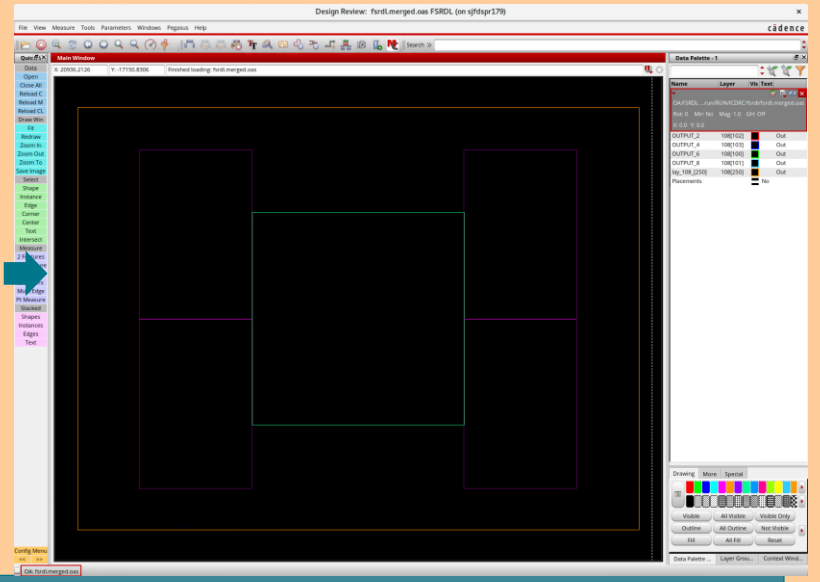
Inter-Chiplet DRC rule deck

Side File for Marker layer projection

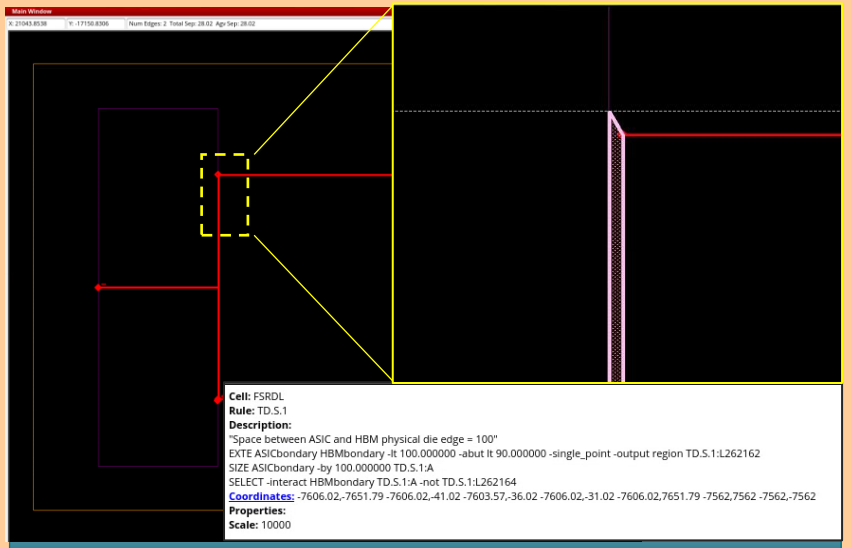
Inter-chiplet DRC check on design GDS with markers

Pegasus™ System

Auto generated GDS with Markers



Generate the design layout with the marker layer shapes for signoff

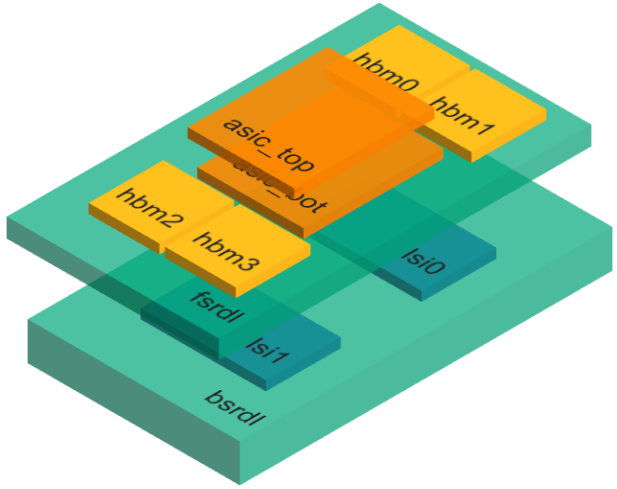


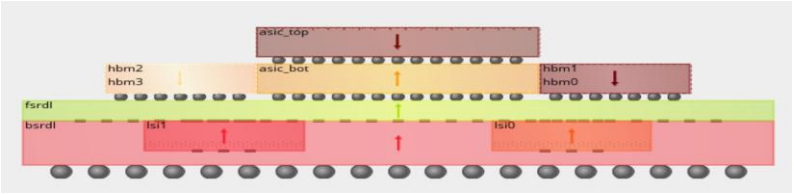
Inter-Chiplet DRC results in Result Viewer

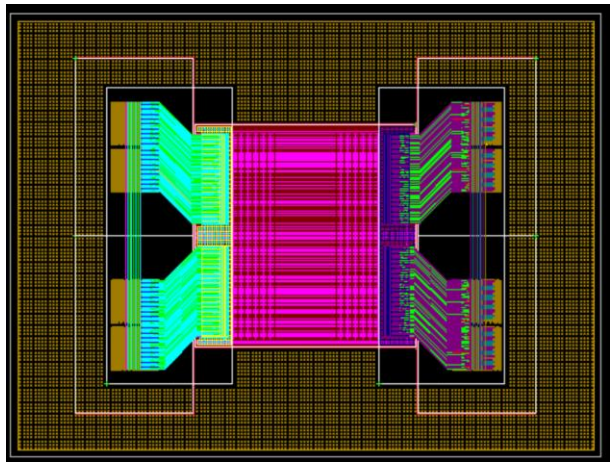
© 2026 Cadence Design Systems, Inc. All rights reserved.

3.5D Design and Analysis RAK

Stacking dies on RDL interposer with silicon bridges







- Integrity™ 3D-IC Platform based 3.5D RAK provides a user end-to-end design methodology for a 3.5D system. Key flows/features included in the RAK:
  - 3.5D full system heterogeneous stack creation
  - System and bump planning
  - 3D stacking dies implementation, optimization, and analysis (more details to follow)
  - Unified Routing Cockpit (URC) to route both silicon bridges and RDLs
  - Early/signoff IR drop analysis
  - Early/signoff thermal analysis
  - Early/signoff system LVS
  - Inter-chiplet DRC verification
  - Signoff SI analysis
- It can serve as a generic Cadence reference flow for 3.5D stacking dies on RDL interposer and silicon bridge style process, such as TSMC CoWoS-L, Intel EMIB, Samsung I-CubeE
- 3.5D RAK early access for selected customers only

© 2026 Cadence Design Systems, Inc. All rights reserved.



© 2026 Cadence Design Systems, Inc. All rights reserved worldwide. Cadence, the Cadence logo, and the other Cadence marks found at https://www.cadence.com/go/trademarks are trademarks or registered trademarks of Cadence Design Systems, Inc. Accellera and SystemC are trademarks of Accellera Systems Initiative Inc. All Arm products are registered trademarks or trademarks of Arm Limited (or its subsidiaries) in the US and/or elsewhere. All MIPI specifications are registered trademarks or service marks owned by MIPI Alliance. All PCI-SIG specifications are registered trademarks or trademarks of PCI-SIG. All other trademarks are the property of their respective owners.